

13-3a Fixed and Variable Costs

Caleb’s total cost can be divided into two types. Some costs, called **fixed costs** (costs that do not vary with the quantity of output produced) , do not vary with the quantity of output produced. They are incurred even if the firm produces nothing at all. Caleb’s fixed costs include any rent he pays because this cost is the same regardless of how much coffee he produces. Similarly, if Caleb needs to hire a full-time bookkeeper to pay bills, regardless of the quantity of coffee produced, the bookkeeper’s salary is a fixed cost. The third column in **Table 2** shows Caleb’s fixed cost, which in this example is **\$3.00**.

Some of the firm’s costs, called **variable costs** (costs that vary with the quantity of output produced) , change as the firm alters the quantity of output produced. Caleb’s variable costs include the cost of coffee beans, milk, sugar, and paper cups: The more cups of coffee Caleb makes, the more of these items he needs to buy. Similarly, if Caleb has to hire more workers to make more cups of coffee, the salaries of these workers are variable costs. Column (4) in the table shows Caleb’s variable cost. The variable cost is 0 if he produces nothing, **\$0.30** if he produces 1 cup of coffee, **\$0.80** if he produces 2 cups, and so on.

A firm’s total cost is the sum of fixed and variable costs. In **Table 2**, total cost in column (2) equals fixed cost in column (3) plus variable cost in column (4).